

TEACHING Exam

Subject – MATHS

Topic – SIMPLIFICATION

LECTURE NO - 01



By– Himanshu Sir

Topics to be Covered

Topic

SIMPLIFICATION



$$\frac{1}{3.5} + \frac{1}{5.7} + \frac{1}{7.9} + \frac{1}{9.11} + \frac{1}{11.13} + \frac{1}{13.15} = ?$$

(1) $\frac{11}{18}$

(2) $\frac{24}{53}$

(3) $\frac{56}{53}$

(4) $\frac{2}{15}$

$$\frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{13 \times 15}$$

$$= \frac{\cancel{6} \times 2}{3 \times 15} = \frac{2}{15}$$

$$\frac{1}{2.5} + \frac{1}{5.8} + \frac{1}{8.11} + \frac{1}{11.14} + \frac{1}{14.17} + \frac{1}{17.20} + \frac{1}{20.23} + \frac{1}{23.26} + \frac{1}{26.29} = ?$$

(1)

$$\frac{8}{9}$$

(2)

$$\frac{24}{53}$$

(3)

$$\frac{56}{53}$$

(4)

$$\frac{9}{58}$$

$$\frac{9}{2 \times 29} = \frac{9}{58}$$

$$\frac{1}{1.5} + \frac{1}{5.9} + \frac{1}{9.13} + \frac{1}{13.17} + \frac{1}{17.21} = ?$$

(1) $\frac{11}{18}$

~~(2)~~

$\frac{5}{21}$

$$\frac{5}{1 \times 21} = \frac{5}{21}$$

(3) $\frac{14}{65}$

(4) $\frac{56}{53}$

$$\frac{1}{4.\underline{7}} + \frac{1}{\underline{7}.10} + \frac{1}{\underline{10}.13} + \frac{1}{13.16} + \frac{1}{16.19} = ?$$

(1) $\frac{11}{19}$

~~(2)~~ $\frac{5}{76}$

(3) $\frac{14}{65}$

(4) $\frac{15}{53}$

$$\frac{5}{4 \times 19} = \frac{5}{76}$$

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} = ?$$

$\downarrow$
 $\textcircled{1} \times 2$ 2×3 3×4 . . . $8 \times \textcircled{9}$

- ~~(1)~~ $\frac{8}{9}$
 (2) $\frac{56}{53}$
 (3) $\frac{24}{53}$
 (4) $\frac{14}{65}$

$$\frac{8}{1 \times 9} = \frac{8}{9}$$

$$\frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{110} + \frac{1}{132} = ?$$

5×6 6×7 11×12
 (1)

~~(2)~~
 $\frac{7}{60}$

(3)
 $\frac{8}{66}$

(4)
 $\frac{14}{65}$

$$\frac{7}{5 \times 12} = \frac{7}{60}$$

$$\frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{110} + \frac{1}{132} + \frac{1}{156} + \frac{1}{182} = ?$$

5×6
 (1) $\frac{9}{70}$
 (2) $\frac{9}{70}$
 (3) $\frac{7}{72}$
 (4) $\frac{9}{72}$

13×14
 $\frac{9}{5 \times 14} = \frac{9}{70}$

$$\frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{110} + \frac{1}{132} + \frac{1}{156} = ?$$

3x4
(1) $\frac{11}{18}$

~~(2)~~ $\frac{10}{39}$

(3) $\frac{14}{65}$

(4) $\frac{56}{53}$

12x13

$$\frac{10}{3 \times 13} = \frac{10}{39}$$



$$\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \frac{1}{4.5.6} = ?$$

$$\frac{\cancel{4} \times 7}{5 \times 6} \times \frac{1}{\cancel{4}}$$

$$= \frac{7}{30}$$

(1) $\frac{11}{18}$

(2) $\frac{24}{53}$

(3) $\frac{56}{53}$

(4) $\frac{7}{30}$

$$\frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \frac{1}{3 \times 4 \times 5} + \frac{1}{4 \times 5 \times 6}$$

$$\frac{1}{6} + \frac{1}{24} + \frac{1}{60} + \frac{1}{120}$$

$$\frac{20 + 5 + 2 + 1}{120} = \frac{28}{120} = \frac{7}{30}$$

$$\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \frac{1}{4.5.6} + \frac{1}{5.6.7} + \frac{1}{\underline{6.7.8}} = ?$$

(1) $\frac{27}{18}$

~~(2)~~ $\frac{27}{112}$

(3) $\frac{56}{53}$

(4) $\frac{7}{30}$

$$= \frac{1}{4} \times \frac{3 \times 6 \times 9}{2 \times 7 \times 8} = \frac{27}{112}$$

$$\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \frac{1}{4.5.6} + \frac{1}{5.6.7} + \frac{1}{6.7.8} + \frac{1}{\underline{7.8.9}} ?$$

(1) $\frac{27}{18}$

(2) $\frac{27}{144}$

~~(3)~~ $\frac{35}{144}$

(4) $\frac{35}{44}$

$$\frac{7 \times 10^5}{8 \times 9} \times \frac{1}{4} = \frac{35}{144}$$

2

$$\textcircled{1} \frac{1}{2} + \textcircled{11} \frac{1}{2} + \textcircled{111} \frac{1}{2} + \textcircled{1111} \frac{1}{2} = ?$$

Max-4 अंक

- (a) 1236
- (b) 1224
- (c) 1220
- (d) None of these

$$1234 + 2 = 1236$$

$$\underline{1}\frac{1}{2} + \underline{11}\frac{1}{2} + 111\underline{\frac{1}{2}} + \underline{1111}\frac{1}{2} + \underline{11111}\frac{1}{2} + \underline{111111}\frac{1}{2} = ?$$

- ☒ (a) 123459
- (b) 123456
- (c) 123458
- (d) None of these

$$1234.56 + 3 = 123459$$

$$2\frac{1}{2} + 22\frac{1}{2} + 222\frac{1}{2} + 2222\frac{1}{2} = ?$$

(a) 2468

☒ (b) 2470

(c) 1272

(d) None of these

$$= \overset{1}{2} + \overset{2}{22} + \overset{3}{222} + \overset{4}{2222} = 2468$$

$$+ 2$$

$$2470$$

$$2(1 + 11 + 111 + 1111)$$

$$1234 \times 2 = 2468 + 2$$

$$= 2470$$

$$3\frac{1}{2} + 33\frac{1}{2} + 333\frac{1}{2} + 3333\frac{1}{2} = ?$$

- (a) 3702
- ☒ (b) 3704
- (c) 3706
- (d) 3708
- $$\begin{array}{ccccccc} & 1 & & 2 & & 3 & & 4 \\ & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \underline{3} & + & \underline{33} & + & \underline{333} & + & \underline{3333} & + 2 \end{array}$$
- $$3702 + 2 = 3704$$

2 minutes Summary

Topic

SIMPLIFICATION



THANK YOU

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